

SUMMARY

This video presents Moore and Mealy machines, emphasizing construction techniques, conversion logic, arithmetic applications, and state minimization principles.

Core insights:

Moore machines associate output with states and produce output length $n+1$ for input length n .

Mealy machines associate output with transitions and produce output length n .

Mealy machines are often more state-efficient for arithmetic tasks.

State minimization depends on identifying equivalent states with identical transitions and outputs.

1. FUNDAMENTAL PRINCIPLES OF MOORE AND MEALY MACHINES

MOORE MACHINES

In a Moore machine, output depends only on the current state.

Key Properties:

Output Length:

Input length = n

Output length = $n + 1$

Reason:

Initial state produces an ϵ output before processing input.

Transition Requirement:

Each state must define transitions for every symbol in the input alphabet (complete system property).

MEALY MACHINES

In a Mealy machine, output depends on both the current state and the input symbol.

Key Properties:

Output Length:

Input length = n
Output length = n

Conversion Principle:

When converting Moore $\rightarrow$ Mealy:

Number of states remains unchanged.
State outputs are transferred to incoming transitions.

2. PRACTICAL MACHINE CONSTRUCTION EXAMPLES

Task Type	Core Logic	Output Configuration
Counting Substring "ab"	Track progress through states representing partial match	Output 1 upon full match; otherwise 0
Consecutive Pattern "aa"	Self-loop on final state to allow overlap	Output 1 at completion of pattern
Binary 1's Complement	Flip bits ($0 \rightarrow 1$, $1 \rightarrow 0$)	Mealy: 1 state; Moore: 2 states
Binary 2's Complement	Keep bits unchanged until first '1' from LSB; flip thereafter	Requires LSB-first processing
Residue Mod n	States represent remainders 0 to $n-1$	Output equals remainder index

3. ADVANCED ARITHMETIC AND MODULO APPLICATIONS

RESIDUE MODULO CONSTRUCTION (AD RULE)

For computing number mod k :

Step 1:

Define states $Q_0, Q_1, \dots, Q_{k-1}$
Each state represents remainder.

Step 2:

Define transitions cyclically based on input.

Example (Binary Mod 3):

State	Input 0	Input 1
Q_0	Q_0	Q_1
Q_1	Q_2	Q_0
Q_2	Q_1	Q_2

Output corresponds to state index.

BINARY SUMMATION MACHINE

Binary addition requires tracking carry.

States:

$Q_0 \rightarrow$ No Carry

$Q_1 \rightarrow$ Carry

Logic:

In Q_0 :

$1 + 1 \rightarrow$ Output 0, move to Q_1

In Q_1 :

$0 + 0 \rightarrow$ Output 1, move to Q_0

Carry propagates until resolved.

4. STATE MINIMIZATION AND OPTIMIZATION

Minimization improves efficiency by merging equivalent states.

Two states are equivalent if:

1. Transition Equality:

For every input symbol, both transition to identical (or equivalent) states.

2. Output Equality:

For every input symbol, both produce identical outputs.

EXAMPLE EQUIVALENCE CASE

State Pair	Reason for Equivalence
B and E	Both transition to D and C with outputs (0,0)
D and H	Identical transitions and outputs
A and C	Become equivalent after first merge

Minimization is iterative until no further merges are possible.

5. ANALYSIS OF GATE PREVIOUS YEAR QUESTIONS

Synchronizing Sequence (1995)

Problem:

Find shortest sequence to reach state C from unknown start.

Solution:

Sequence "10"

Reason:

Regardless of starting state, input "10" always leads to state C.

Key Insight:

Must work for all possible initial states.

2's Complement Machine (2002)

Observation:

Machine processes from LSB.

Keeps bits unchanged until first '1'.

Flips all bits thereafter.

Conclusion:

Standard 2's complement calculator.

Minimum Input Length (2009)

Objective:

Move from state 00 to state 01 with output 1.

Observation:

Direct transition gives output 0.

Shortest valid path requires 3-bit sequence.

Answer:

"101"

6. KEY DEFINITIONS

Term	Definition
LSB	Least Significant Bit (rightmost bit)
Unreachable State	State not reachable from initial state
ϵ Output (Moore)	Output of initial state before input
Residue Modulo	Remainder after division